

Microbial association networks

Introduction to Metagenomics

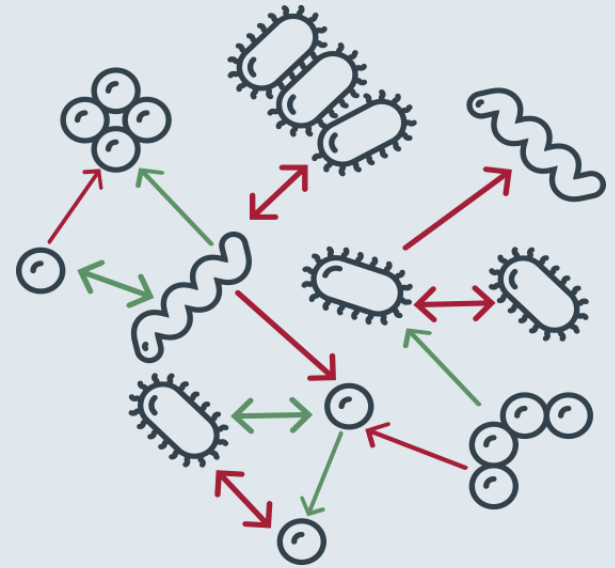
Sebastián Ayala Ruano

Data Science Platform (DSP)

DTU-Biosustain

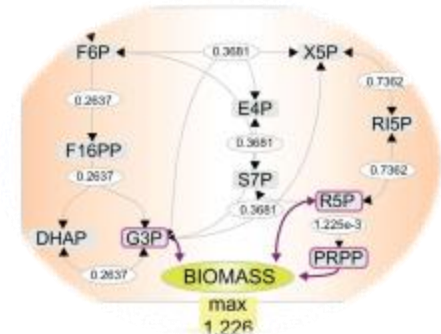
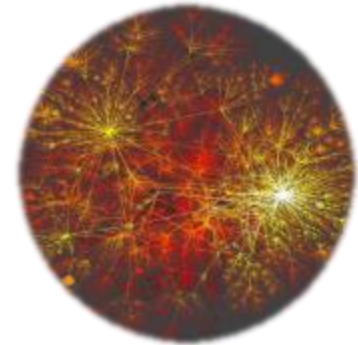
Spring Courses

June 11, 2025



What is a Network?

- Data structure of **components (nodes)** connected by **relationships (edges)**
- **Broad concept** that describes how things interact to each other
- Networks are everywhere



Why Networks?

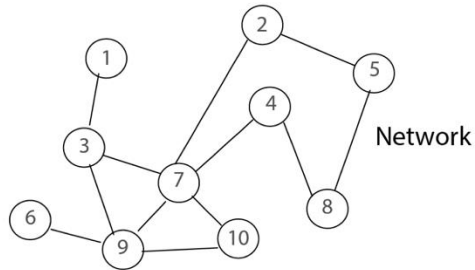
These structures allow:

- Quick **integration** of **heterogeneous data** based on relationships
- **Graph theory** methods can be used to **analyze** and **interpret** data, e.g., topological properties can be used to explain:
 - The possible **role** of specific components
 - The identification of **clusters** of nodes
 - The **flow** of information
 - The **robustness** of the system
- Examine **similarities and differences** between systems
- **Visualize** data – complex systems can be represented as networks



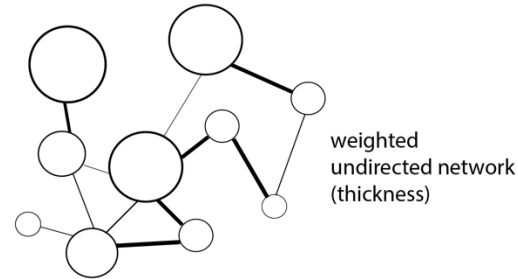
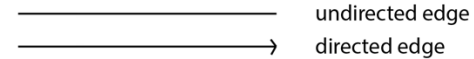
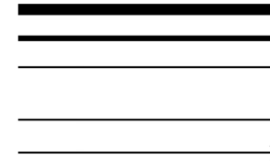
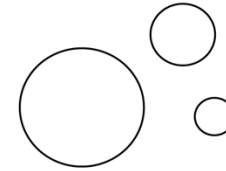
Basic Network Concepts

$$G = (V, E)$$



0	0	1	0	0	0	0	0	0	0
0	0	0	0	1	0	1	0	0	0
1	0	0	0	0	1	0	1	0	0
0	0	0	0	0	0	1	1	0	0
0	1	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	1	0
0	1	1	1	0	0	0	0	1	1
0	0	0	1	1	0	0	0	0	0
0	0	1	0	0	1	1	0	0	1
0	0	0	0	0	0	1	0	1	0

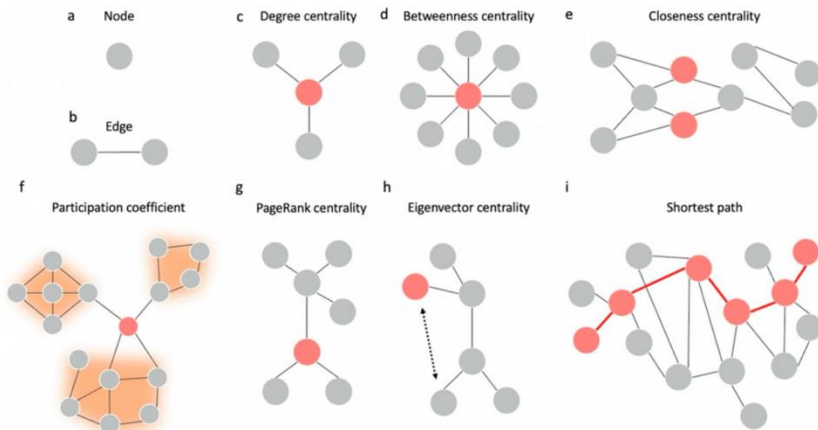
Adjacency matrix



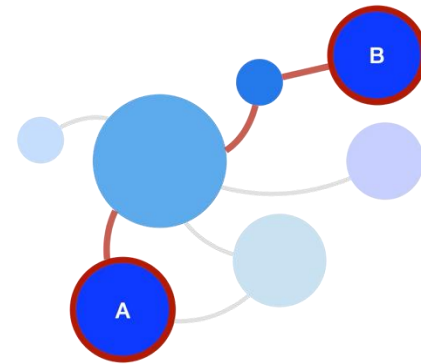
0	0	0	0	0	0	0
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Weighted adjacency matrix

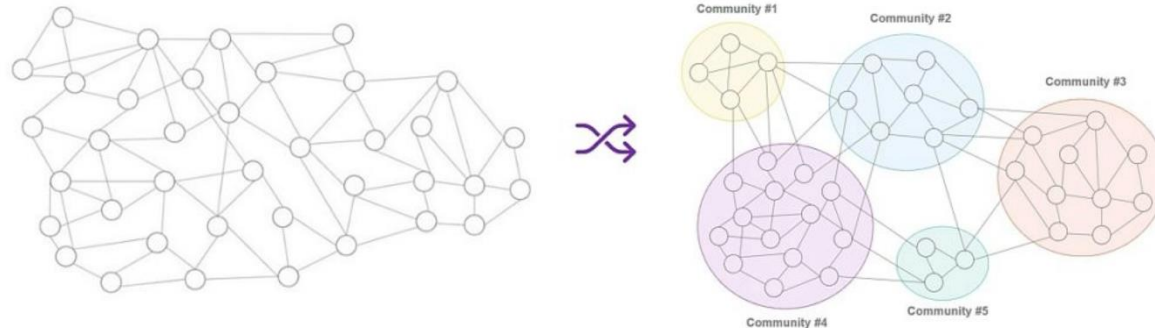
Basic Network Concepts – Topological Properties



Centrality



Information Flow



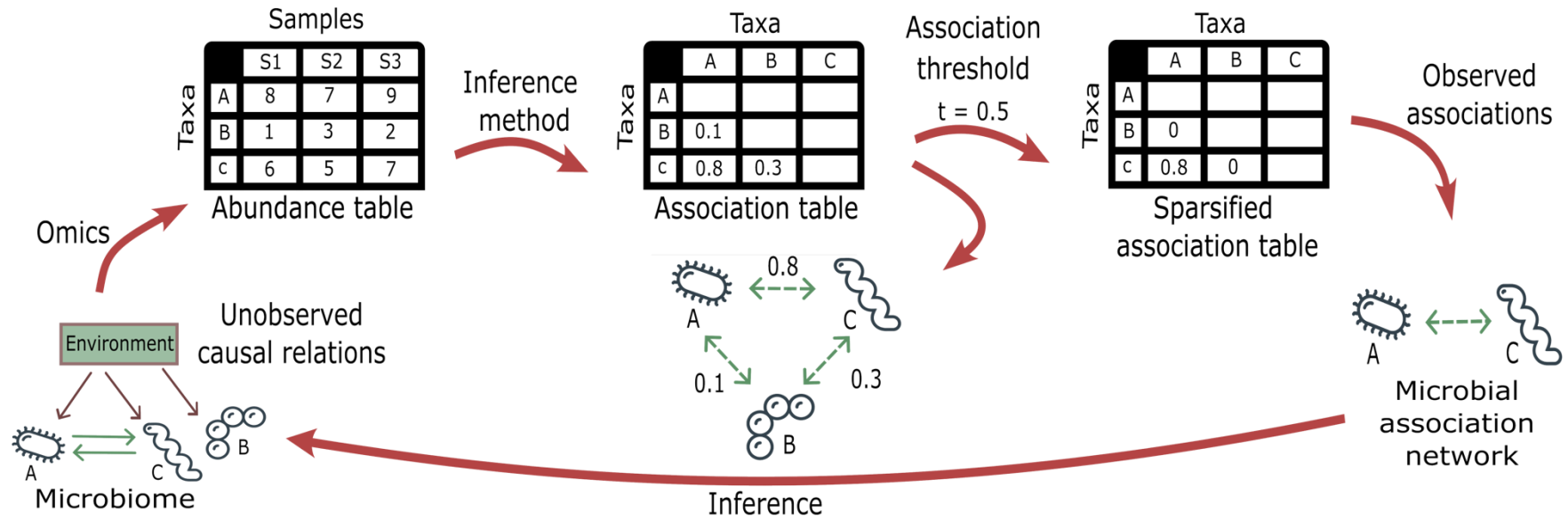
Clustering

Tanglay, et al., 2023, Cancers

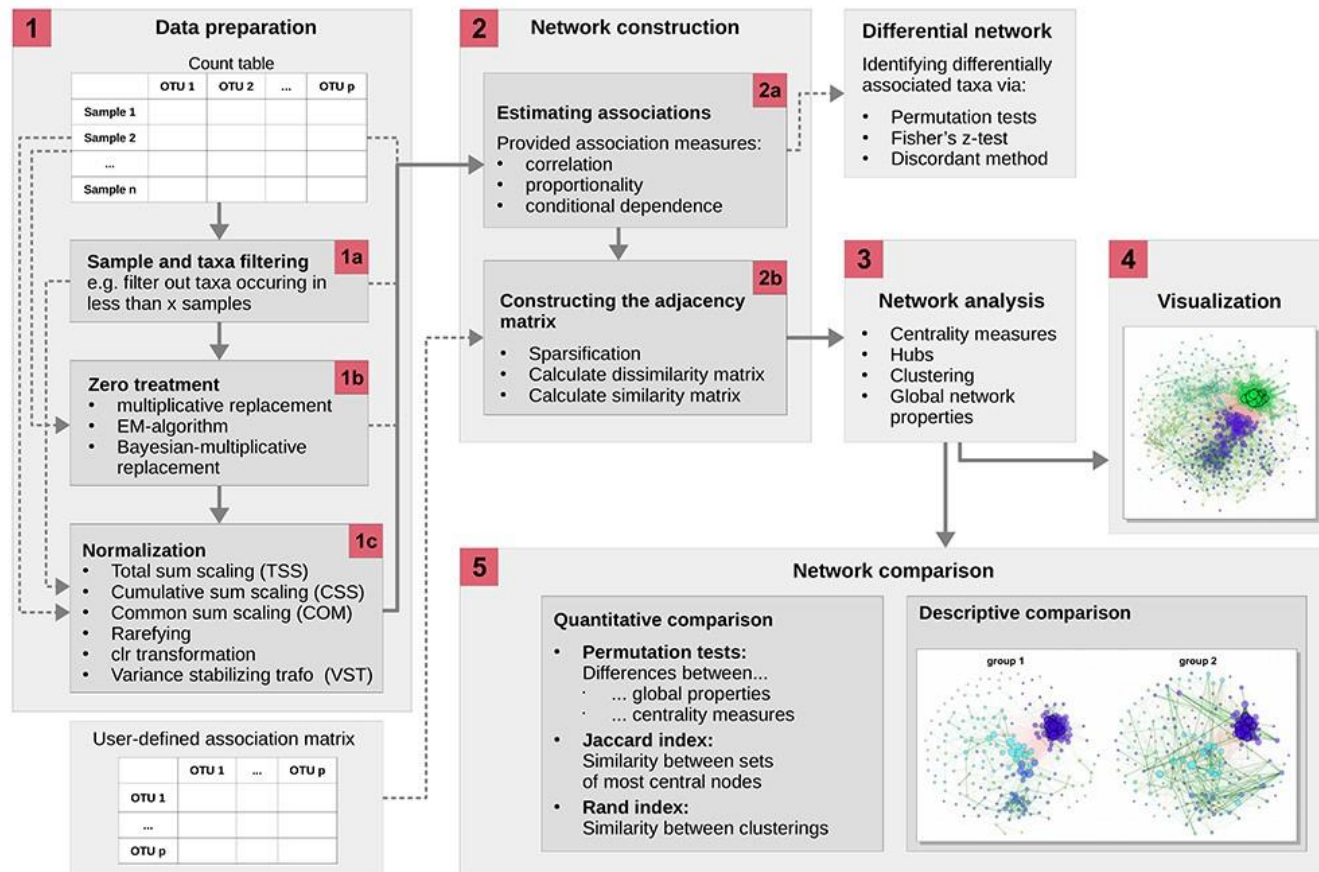
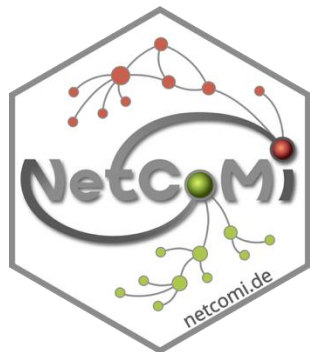
<https://timbr.ai/community-detection-algorithm/>

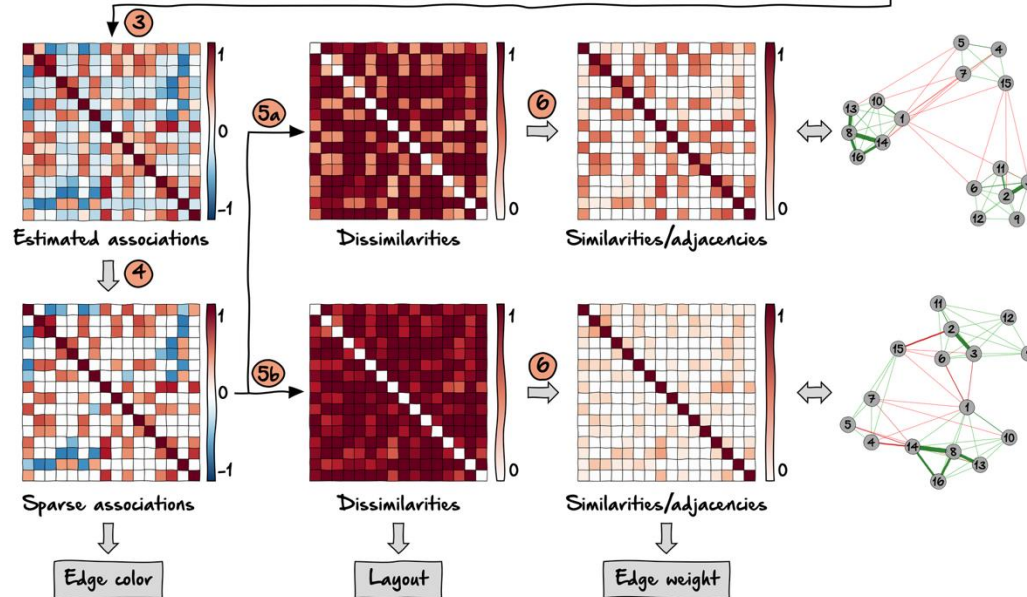
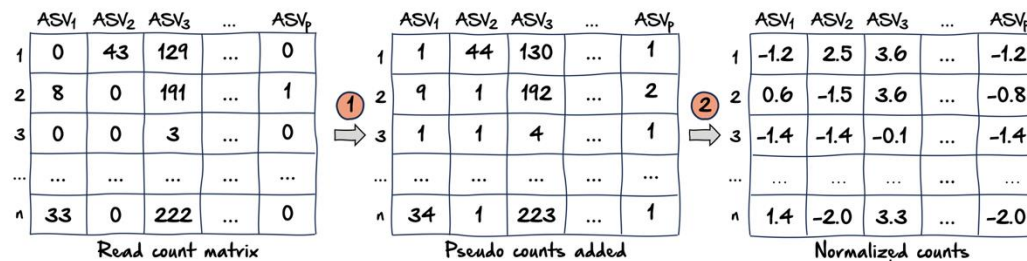
Microbial association networks (MANs)

- **MAN:** weighted and undirected network, where nodes are microbes and edges indicate co-presence or mutual exclusion abundance patterns across samples
- Weights are association values among microbes defined by an inference method

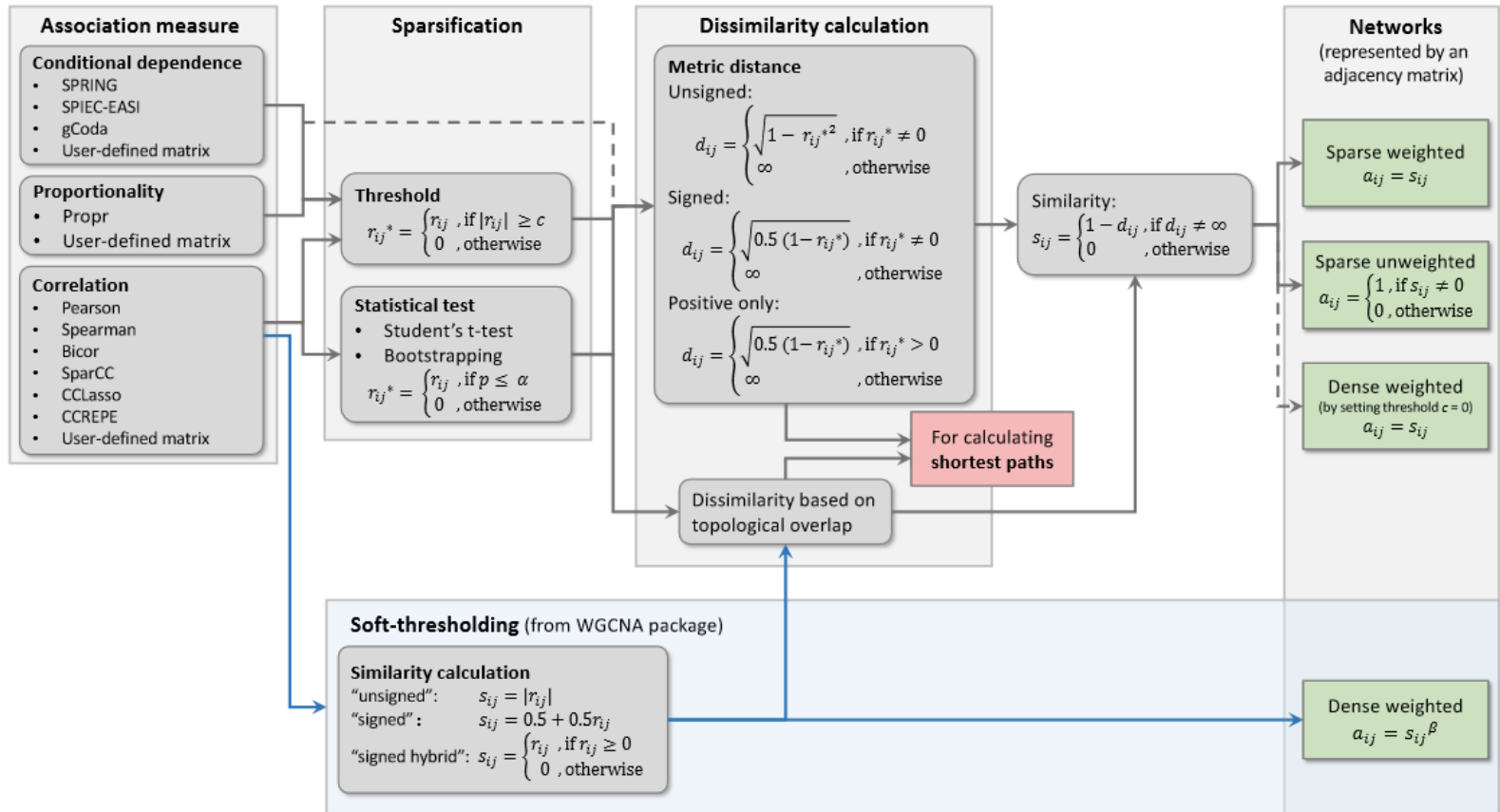
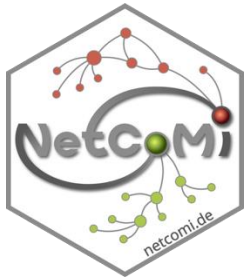


NetCoMi – General Workflow





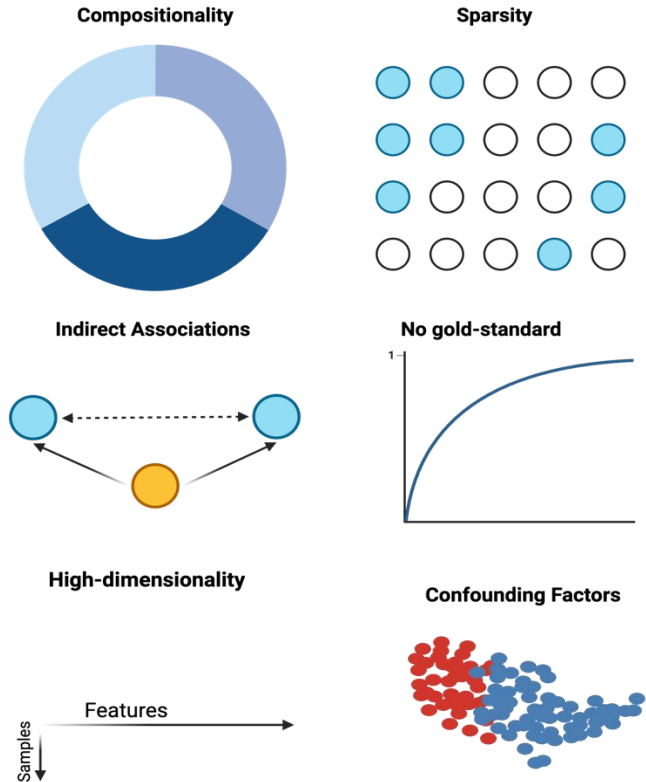
NetCoMi – Network Types



CCLasso

Limitations and Challenges of MANs

- **Compositionality of microbial data** — only **relative abundances** are measured, not absolute counts
- **Sparsity of data** — containing **many zero counts** due to low abundance or detection limits
- **High-dimensionality with small sample size** — Often the number of microbial taxa (**features**) greatly **exceeds** the number of samples (**observations**)
- **Indirect associations** — hard to **distinguish** actual **interactions** from **secondary effects**
- **Lack of ground truth** — no “gold standard” network to validate inferred MANs
- **Confounding factors** — **host factors** (e.g., diet, health, medications) or **technical artefacts** (e.g., batch effects, depth)



Do you want to know more?

- **Course GitHub repo:** https://github.com/biosustain/dsp_metagenomics_training
- **Docs:** https://biosustain.github.io/dsp_metagenomics_training
- **NetCoMi:** <https://netcomi.de/>
- **MANs:** https://microbiome.github.io/OMA/docs/devel/pages/network_learning

References

- Layeghifard, M., Hwang, D. M., & Guttman, D. S. (2017). Disentangling Interactions in the Microbiome: A Network Perspective. *Trends in Microbiology*, 25(3), 217–228
- Peschel, S., et al. (2021). NetCoMi: network construction and comparison for microbiome data in R. *Briefings in bioinformatics*, 22(4), bbaa290
- Tanglay, O., et al. (2023). Graph theory measures and their application to neurosurgical eloquence. *Cancers*, 15(2), 556